

LAB WORKSHEET

Lab 18 — Data at Rest (LUKS) and Data in Transit (TLS / SSH)

Name	_____	Date	_____
------	-------	------	-------

Objective: 3.1 — Data security: encryption at rest/in transit, BIOS/bootloader passwords, retention.

Work through the lab on the Killercoda Ubuntu playground and record your results below.

Task 1 — LUKS lifecycle

Format, open, write, close and verify the LUKS volume.

Paste the 'no plaintext visible' result.

Task 2 — Wrong passphrase

Try to open with a wrong passphrase.

Record that it was rejected.

Task 3 — TLS cipher

Serve HTTPS and inspect the negotiated protocol/cipher.

Record the Protocol and Cipher lines.

Task 4 — SSH hardening

List the sshd_config hardening lines.

Write the five key hardening directives.

Task 5 — Retention

Map retention/destruction to four classifications.

Fill the classification table.

Completion checklist

- Completed the full LUKS at-rest lifecycle and verified ciphertext.
- Served TLS and inspected the negotiated cipher.
- Listed the SSH hardening directives.
- Explained BIOS/UEFI and GRUB password roles.
- Mapped retention to data classification.